Reading: Computer Components and Functions

a. System software: Software used to manage computer resources, for example, the CPU, Memory, I/O devices, etc., fall under the category of System Software. Examples are operating systems, editors, compilers, and assemblers.

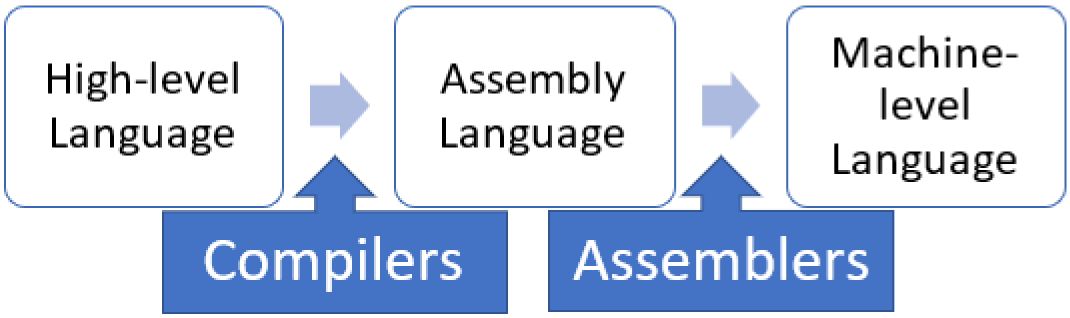
b. Application software: Software used to perform general or specific tasks inside a computer. Hence can be categorized into general purpose and specific purpose application software.

Computer Language:

A computer is a digital device in which a processor understands instructions written in the form of a binary code i.e., any instruction or number using only 0’s and 1’s combination. Software written in the form of binary code is said to use Machine Level Language.

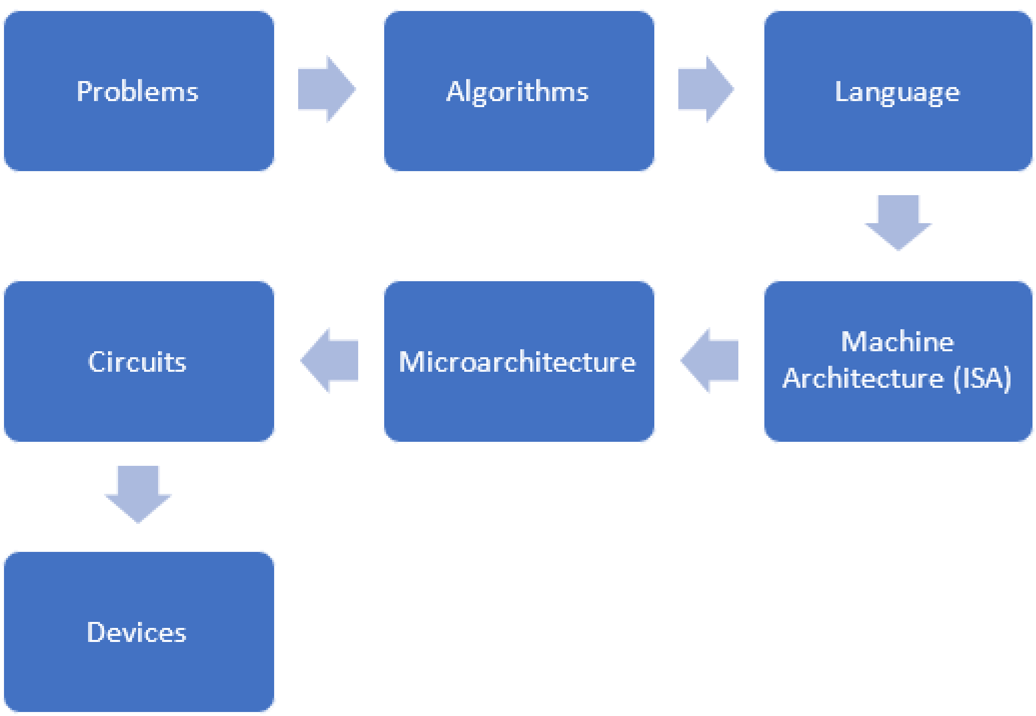
In general, it’s quite difficult to write software programs in binary code. So, software developers use a mnemonic-based language called the Assembly Language. Assembly language enables writing the instruction in a simpler form which is further converted into Machine Language using a special tool called the Assembler.

Nowadays, we use High-Level Languages to develop software, which is similar to using the English language. Some examples of high-level languages are C, C++, Java, etc. Before execution by the computer, this high-level language software program is converted to Assemble Language using a special tool called the Compiler.



Levels of Transformation in Computer Systems:

Before a computer can solve a problem, different transformations happen across multiple levels, as shown below:



- Problems are generally specified in natural languages, like English, French, German, or other spoken languages that humans use to converse. Unfortunately, this is not a language that computers understand.

- Hence, the first transformation happens when we convert our problem statement into an Algorithm. An algorithm is a step-by-step procedure that a computer can carry out.
- Once the algorithm is selected, the next step is to translate it into a computer program using a high-level language since it is simpler to code and easier to debug for any errors. Also, because high-level languages are machine independent, it enables the program to be compiled for different machines.
- Using compilers, the next transformation happens when a software program written using a particular programming language, is converted into machine instructions that the processor can understand.
- The next transformation level involves Microarchitectures, which involves implementing the computer's Instruction Set Architecture (ISA).
- Once the microarchitecture of a computer is designed, we need to implement each sub-part of this design. Each element of the microarchitecture gets implemented using simple logic circuits, which comprise the basic building blocks called the devices.

History of Computers:

Charles Babbage invented the first computer. He was born in December 1791 and died aged 79 years in Oct 1871. He is well-known as a mathematician, philosopher, engineer, and inventor. But most importantly, he is well-known as the father of the computer. Here is the further generation-wise development of computers.

Generation Name	**Machine Name**

Generation – 0 (1623 – 1945)	Mechanical Calculating Machines • Wilhelm Schikard – “Calculating Clock” • Blaise Pascal – “Pascaline” • “Lightning Portable adder” and “Addometer”
Generation – 1 (1943 – 1953) **(Implemented using Vacuum Tubes)**	Vacuum Tubes Computers, EDVAC, ENIAC
Generation – 2 (1954 – 1965) **(Implemented using Transistor)**	IBM 7000 and DEC-PDP1
Generation – 3 (1965 – 1980) **(Used Integrated Circuits)**	IBM 360
Generation – 4 (1980 Onwards) **(Based on VLSI Technology)**	Apple 1, Apple 2, IBM PC

Advances in Semiconductor Technology:

Based on the number of devices that the IC can accommodate, semiconductors can be classified as follows:

Technology	Number of Devices on a Chip
Small-scale integration (SSI)	1-100 devices

Medium-scale integration (MSI)	100-3,000 devices
Large-scale integration (LSI)	3,000 - 100,000 devices
Very large-scale integration (VLSI)	100,000 - 100,000,000 devices
Ultra large-scale integration (ULSI)	Over 100,000,000 devices

Little Computer 3 (LC-3):

The Little Computer 3 (LC-3) was developed jointly by Yale N. Patt at the University of Texas at Austin and Sanjay J. Patel at the University of Illinois at Urbana–Champaign. It includes the most important features of the well-known computer systems available in the market. The instruction set of LC-3 has only 15 instructions identified by unique codes. These represent the different types of arithmetic, logic, and control operations that the LC-3 can perform.

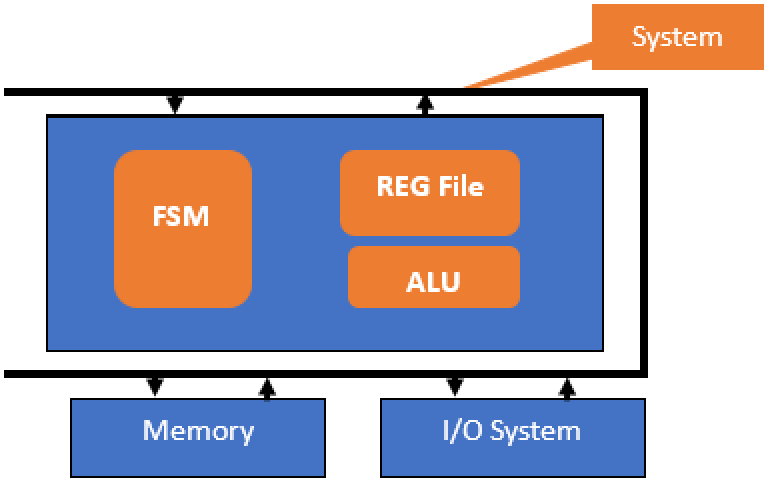
- **LC-3 Architecture**:

The three important parts of LC-3 are memory, the Input and Output System, and the Control and Processing Unit (CPU). The Processor Bus or the System Bus connects the different parts, transferring information to and from each component. Here, the CPU consists of the following key blocks:

1) Finite State Machine (FSM),

2) Arithmetic and Logic Unit (ALU), and

3) Register File or the Register Set.



- **LC-3 Simulator**:

To write and execute the program in LC-3, we do not require to purchase any hardware-based LC-3 computer. A simulator called LC-3 simulator, is available in Windows and Linux Machines, as well as a web version. Here, every instruction stored in the LC-3 memory can be specified by:

1) Giving the value of that instruction and storing it in the correct memory location.

2) Providing the raw instruction code in text form.

3) Writing assembly code in the text editor and load to the simulator.

Common Terms Used for Measurement

- Kilo- (K) = 1 thousand = 10^3 and 2^{10}
- Mega- (M) = 1 million = 10^6 and 2^{20}
- Giga- (G) = 1 billion = 10^9 and 2^{30}
- Tera- (T) = 1 trillion = 10^{12} and 2^{40}
- Peta- (P) = 1 quadrillion = 10^{15} and 2^{50}

Measurement of Speed

The base unit for the measurement of speed is Hz. Hertz stands for clock cycles per second, which is also a unit of frequency. So, when we measure the speed of operation of any component in the computer system, we look at the number of processing cycles the component can execute per second time.

Measurement of Storage

The basic unit of storage in a computer system is a byte. A byte consists of 8 bits, where each bit can take two values - a 0 or a 1. Kilo Bytes or Mega Bytes or Giga Bytes are other units to measure the capacity of storage units. The main memory in a computer system, also called the RAM or the Random Access Memory, is usually measured in Mega Bytes. On the other hand, disk storage is measured in Giga Bytes for small systems and in Tera Bytes for large systems.

Measurement of Time

- Milli- (m) = 1 thousandth = 10^{-3}
- Micro- (µ) = 1 millionth = 10^{-6}
- Nano- (n) = 1 billionth = 10^{-9}
- Pico- (p) = 1 trillionth = 10^{-12}
- Femto- (f) = 1 quadrillionth = 10^{-15}

Reading Summary:

In this reading, you have learned the following:

- The need for a computer system
- Hardware and software components of a computer
- The levels of abstraction in computing systems
- The most significant events and concepts in the history of computers
- The high-level architecture of the LC-3 computer and use the LC-3 Simulator